

Lab # 10

SEMAPHORES

Objective

Study the features of Semaphores.

Exercises

1. What was the mistake in the Program 1, and how it is rectified in Program 2?

The mistake in Program 1 was that it initialized `nsem` to 0, but a semaphore set must contain at least one semaphore, so `semget ( )` returned an error. This was rectified in Program 2 by changing `nsem` to 1, which creates a valid semaphore set and allows `semget ( )` to successfully create the semaphore.

2. Do `ipcs -s` at the prompt to see a listing of the semaphores and an entry for the semaphore will be displayed.

```
stewiedev@DESKTOP-RH97L4T:~$ ipcs -s
----- Semaphore Arrays -----
key          semid      owner       perms      nsems
0x00000000  0          stewiedev   666        1
0x00000000  1          stewiedev   666        1
0x00000000  2          stewiedev   666        1
```

3. Run the Programs 3 and 4, and find out what is the problem in 3 and how it is rectified in 4.

The problem in Program 3 is that it only retrieves the PID using `semctl ( )` with `GETPID`, but it does not perform any operation that sets or updates the semaphore PID value. As a result, it may return an old or invalid PID. Program 4 rectifies this by performing a semaphore operation first (such as using `semop ( )`), which updates the semaphore information. After that, `GETPID` correctly returns the PID of the process that last accessed the semaphore.